Inverse rendering through a multiphysics equilibrium

A differentiable LWIR camera composed with a live Fortran/JAX/PyTorch cooling model

il-miscusi | github.com/il-miscusi/tesseract-inverse-thermography

Abstract. A thermal image is not temperature. It is band-integrated Planck radiance, modified by emissivity, reflected ambient, projection, optical blur, vignetting, and sensor calibration. We implement that measurement model as a differentiable JAX Tesseract and pull a pixel loss through it, through a coupled flow-heat fixed point, and back to an unknown volumetric heat source. The coupled system spans a PyTorch viscosity closure, a Fortran Darcy/Brinkman solver with a hand-derived adjoint, and JAX heat transport. The complete pixels-to-source derivative matches finite differences at 3.61e-07 in-process and 1.14e-07 through four served containers. On a frozen 12-scene bank with 64x32 truth, 32x16 inversion, and a held-out camera view, the calibrated system produces useful diagnoses on 12/12 scenes; a 4% emissivity error increases power error on 12/12 pairs by a median 4.42 percentage points (95% bootstrap interval 3.28-4.88). The stronger preregistered harm-prevalence claim is not accepted.

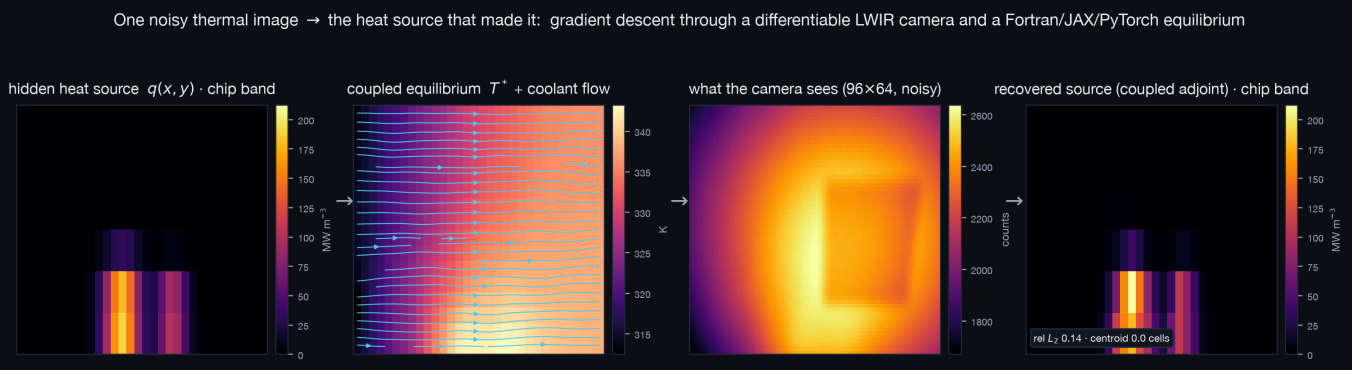

Figure 1. One noisy LWIR frame is inverted through the camera and the coupled equilibrium to recover the hidden heat-source map. Source panels share a physical scale.

Three contributions

Composition. Four typed Tesseracts expose apply and VJP endpoints across three native derivative regimes. A matrix-free implicit adjoint composes them without a global tape.

Evidence. A frozen 12-scene bank uses 64x32 truth, 32x16 inversion, two training views, one held-out view, absolute residual gates, and no excluded failures.

Reproducibility. Fast CI verifies the API path. A separate target builds and serves all four images and stores image IDs, source commit, timing, and the finite-difference table.

Honest limits. Same-grid accuracy, optimizer limits, synthetic data, and the no-PSF negative result are all reported. Claims are bounded to what the artifacts establish.

Evidence dashboard

API gradient	4-container gradient	useful unseen scenes	paired power bias
3.61e-07	1.14e-07	12/12	+4.42 pp

Track 05 thesis. The decisive result is not merely that a camera can be differentiated. Tesseract makes its VJP operationally composable with an implicit cross-language equilibrium, and the frozen bank tests the resulting inverse system beyond its development scenes. The negative preregistered verdict remains visible alongside the positive paired evidence.

One chain rule across four components

The source q drives a steady heat solve. Temperature changes viscosity; viscosity changes flow; flow changes temperature. At equilibrium T* = G(T*; q). The camera maps T* to digital counts C = R(T*). For a pixel loss J, the camera VJP supplies dJ/dT*. The implicit adjoint solves (I - dG/dT)^T lambda = dJ/dT* with GMRES, then one heat VJP returns dJ/dq. Each matrix-vector product is a chain of component VJPs; no framework traces the whole program.

The gradient no single AD framework can trace

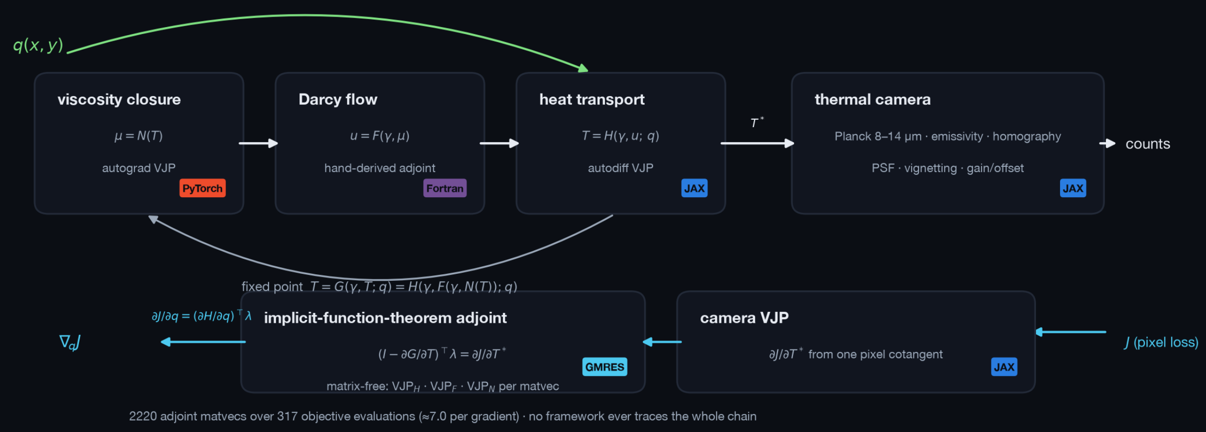

Figure 2. Forward fixed point and reverse implicit adjoint. The fast path uses identical APIs in-process; the interoperability gate serves the four images.

Thermal-camera Tesseract

The JAX renderer integrates Planck spectral radiance over 8-14 um with 64-node Gauss-Legendre quadrature. A grey opaque surface emits eps L(T) and reflects (1-eps)L(T_ambient). A homography and bilinear sampler project radiance to the sensor; a differentiable Gaussian PSF, cos^4 vignetting, gain, and offset complete the image.

Heterogeneous physics

Viscosity is a learned PyTorch closure. Flow is a compiled Fortran Darcy/Brinkman solver with a hand-derived discrete adjoint. Heat transport is JAX with an implicit derivative. Only arrays and typed derivative requests cross component boundaries.

Verification hierarchy

gate	result
camera tests	18 pass
inverse utilities	7 pass
pixels-to-q FD, API	3.61e-07
pixels-to-q FD, 4 containers	1.14e-07
served runtime	80 s

Why Tesseract is load-bearing

JAX cannot trace the PyTorch tape or the Fortran binary; neither tape contains the equilibrium solve. A custom callback layer or rewrite could expose the same mathematical derivative. Tesseract supplies a typed, remotely executable composition without rewriting the native solvers.

Generalization on twelve unseen fault scenes

Experiment D seals seeds 101-112 until the inverse method, two training views, held-out third view, 4% emissivity mismatch, absolute residual gates, and reporting rule are fixed. Truth uses a 64x32 coupled solve and inversion uses 32x16. A known uniform load plus one perturbation at each of 20 chip cells independently calibrates a differentiable observation-space multi-fidelity tangent; no fault scene contributes to it.

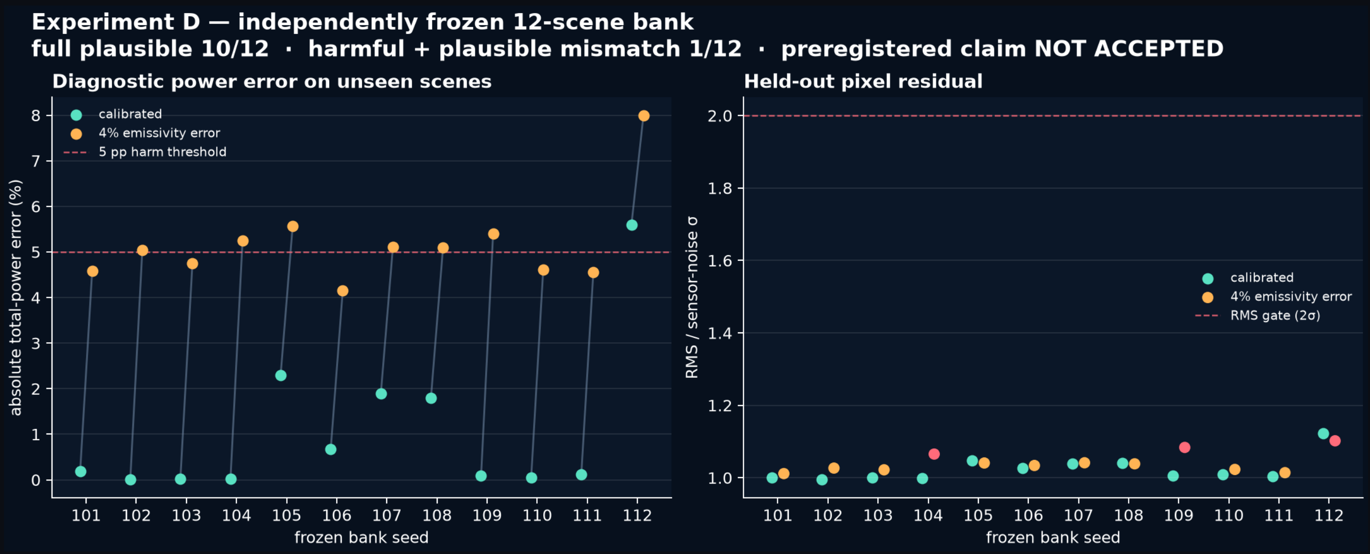

Figure 3. Every paired point is retained. Red mismatch markers fail at least one independent plausibility condition (residual mean or whiteness can reject an arm even when RMS is below 2 sigma).

arm	plausible	useful	RMS counts	centroid mm	power error
calibrated	10/12	12/12	2.014	0.088	0.153%
4% emissivity error	9/12	12/12	2.072	0.087	5.067%

Positive evidence. The calibrated arm is useful on 12/12 scenes. Emissivity mismatch raises absolute power error on 12/12 pairs by a median **4.42 points**; the deterministic paired-bootstrap 95% interval is 3.28-4.88 points.

Negative verdict retained. The stronger preregistered risk claim required at least 6/12 mismatches to be both plausible and at least five points harmful. Only **1/12** qualify, so that claim is **not accepted**.

Why this bank is hard to game

Sealed scenes. Development uses seeds 0-2; the reported bank is exactly 101-112. The summarizer refuses missing or extra seed artifacts and uses a fixed bootstrap seed.

Absolute validation. A held-out fit must satisfy RMS <=4 counts, |mean| <=0.5 counts, and both lag-1 residual correlations <=0.10. Closeness to the calibrated arm is never a plausibility test.

Representability audit. The original exact-source coarse oracle missed fine-grid pixels by 51 counts. Known-load calibration reduced the development oracle to about 2 counts before the bank was opened; the source was not optimized to conceal the discrepancy.

Physical diagnosis. Gates use millimetres and total power, not only source-vector L2. Every optimizer converged, all raw JSON/NPZ pairs and logs are committed, and the final animation uses a real bank trajectory.

What the complete evidence establishes

Forward x gradient factorial

Experiment E separates Experiment B's two interventions. Exact coupled forward/gradient reaches source L2 0.1370 and data loss 2.022. Keeping the coupled forward but truncating only the implicit reverse feedback converges at L2 0.2415 and loss 2.079. Frozen forward/matching gradient reaches L2 0.2457 and loss 3.339. Forward fidelity dominates pixel fit; exact gradient fidelity improves source shape on this fixed problem.

Historical stress test

Experiment C first exposed the inverse crime: with 64x32 truth and 32x16 inversion, calibrated source L2 rose to 1.248 and pixel RMS to 13.95 counts. Its relative calibration-shift gate passed, but the absolute noise-level gate failed. Experiment D repairs representability with independent known-load calibration rather than hiding discrepancy in the source.

Limitations

Measurements remain synthetic; the chip footprint is known; calibration uncertainty is represented by one fixed mismatch rather than a posterior; and material/camera physics are prototype-grade rather than instrument-grade. Real calibrated imagery, transient data, unknown supports, and posterior uncertainty maps are the next validation layer.

Reproduce

```
$ pip install -r requirements.txt
$ make judge
$ make verify
$ make verify-containers
$ make summarize-experiment-d
$ make animation
$ make paper landing
```

Container receipt

Mode: **served_containers**
Best FD error: **1.14e-07**
Runtime: 80 s
Source: 4031a3890dbf
Four immutable image IDs are stored in the JSON artifact.

Artifact map

Frozen protocol: writeup/EXPERIMENT_D_PROTOCOL.md
Bank summary: figures/experiment_d/experiment_d_summary.json
Twelve raw JSON/NPZ pairs: figures/experiment_d/bank/
Container receipt: figures/container_e2e_gradient_check.json
Code: github.com/il-miscusi/tesseract-inverse-thermography

Claim-to-artifact matrix

claim	direct evidence	boundary
Cross-framework derivative is correct	two end-to-end FD receipts	directional checks, small grids
Composed inverse generalizes	Experiment D, 12/12 useful	synthetic, known support
4% emissivity biases inferred power	12/12 pairs; +4.42 pp median	harm prevalence claim failed
Coupled model improves same-grid recovery	Experiment B v2, 0.137 vs 0.246 L2	both arms hit budget
Exact adjoint improves source shape	factorial: 0.137 vs 0.241 L2	one fixed problem

Transfer applications

Electronics cooling. Localize a failing die or blocked channel from the visible cold-plate surface while respecting convection and radiometry.

Additive manufacturing. Pull pyrometer residuals through a differentiable sensor and melt-pool equilibrium to infer hidden heat input.

Active thermography. Infer delamination or subsurface defects from transient surface radiation with a swappable camera component.

References

[1] Pasteur Labs, *Tesseract: universal, autodiff-native software components*, 2026. [2] J. Bradbury et al., *JAX: composable transformations of Python+NumPy programs*, 2018. [3] A. Paszke et al., *PyTorch: An Imperative Style, High-Performance Deep Learning Library*, NeurIPS 2019. [4] P. Virtanen et al., *SciPy 1.0: Fundamental Algorithms for Scientific Computing in Python*, Nature Methods 2020. [5] M. Planck, *On the Law of Distribution of Energy in the Normal Spectrum*, 1901.